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Project 1: Colorizing the Prokudin-Gorskii Photo Collection

Approach

First, I followed the structure of our skeleton code to read in an image, convert to double, compute height, and then separate an image into the BGR color channels. From there, I initially tried to implement the iterative approach for aligning the color plates, so I defined a function to loop over every possible shift within a range of -15 to 15 pixels in both x and y directions. For each shift, I then calculated L2 norm between the reference channel (blue) and the shifted channel (green or red) as my scoring metric and kept track of the lowest L2 norm score, and the corresponding shift with the best score while iterating. Additionally, I know on Ed there were some suggestions about cropping images to improve alignment that were also mentioned, so in my algorithm I cropped around 10% of the borders of each image, although I think there may be a better way to do this based on calculating image overlap? Definitely a future point of interest! I included a side-by-side comparison to highlight the effect of cropping the images below

Single Scale Improvements

Below you can see the difference between my single scale algorithm before + after introducing cropping

Cathedral - Single Scale



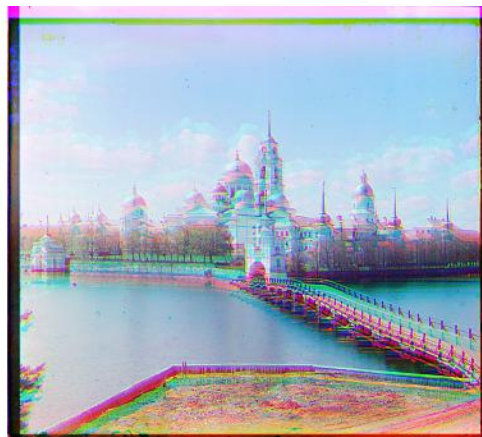
Initial attempt

Cathedral - Single Scale Cropped



Using cropping

Monastery - Single Scale



Initial attempt

Monastery - Single Scale Cropped



Using cropping

Tobolsk - Single Scale



Initial attempt

Tobolsk - Single Scale Cropped



Using cropping

All that being said, this iterative strategy isn't very efficient especially for the larger .tif images, so I decided to implement the image pyramid approach instead.

For this approach, I tried scaling down the input image by half over the course of 3 levels, excluding one level for the original image. Next, starting from the smallest image, I drew on the iterative approach to find the best shift for the green and red channels relative to the blue channel. However, instead of searching over the full range of -15 to 15 pixels, I only searched within a smaller range of -2 to 2 pixels, since the images were smaller. After finding the best shift for each channel at the smallest level, I then scaled up the shifts by a factor of 2 to use as an initial guess for the next level. This process was repeated until reaching the original image size, where I again searched over a range of -15 to 15 pixels to find the final best shifts (using L2 norm as the evaluation metric). Interestingly, I also tried switching out my L2 metric to use NCC, but I didn't really see much improvement, so I think I may need to change some additional parameters or implementations as well. Regardless, the best L2 norm scores and best shifts were updated similarly to the iterative strategy, which is how I ultimately aligned the images and displayed the final colorized image.

Feel free to look through the photos I aligned below!

Overview

Overall, this was a super insightful project! At first, I didn't really understand conceptually how we could colorize images based on these filtered color plates nor could I fathom how Prokudin-Gorskii was able to show the photos in color using different spotlights (I was amazed to read about his technique on the LOC website), but after getting hands-on experience with implementing an algorithmic version of his technique, I feel that I learned a lot not only just about image pyramid strategies but also about photography in general!

I also really loved the variety of the photos! Personally, I love portrait photography, so I was drawn to Prokudin-Gorskii's photos of people! When browsing the LOC collection, I filtered for "Group Portraits", and I was surprised by how many photos of ordinary people there were, although I suppose this could also make sense intuitively because of his methodology and travels. Super valuable to see a snapshot into the lives of all the different people back then, even for just a moment!

CS 180 Gallery

Cathedral



Green: (49, 24), Red: (103, 57)

Church



Green: (25, 4), Red: (58, -4)

Emir



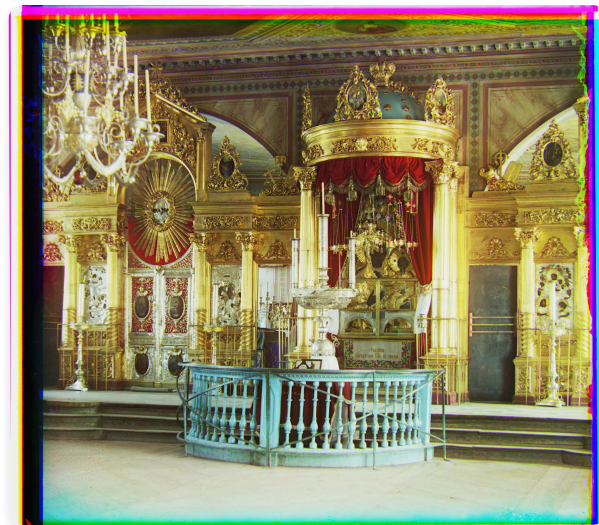
Green: (38, 21), Red: (76, 35)

Harvesters



Green: (53, 14), Red: (112, 11)

Icon



Green: (41, -16), Red: (93, -29)

Italil



Green: (82, 11), Red: (178, 13)

Lastochikino



Green: (-3, -2), Red: (75, -9)

Lugano



Green: (3, 3), Red: (6, 3)

Melons



Green: (41, 17), Red: (90, 23)

Monastery



Green: (-3, 2), Red: (3, 2)

Self Portrait



Green: (5, 2), Red: (12, 3)

Siren



Green: (49, -6), Red: (96, -25)

Three Generations



Green: (79, 29), Red: (176, 36)

Tobolsk



Green: (59, 16), Red: (124, 13)

Additional Collection Photos

Peasant Girls



Green: (-16, 10), Red: (11, 17)

Students



Green: (76, -3), Red: (161, -20)

Tipy Dagestana (Group Photo)



Green: (16, 1), Red: (104, -1)